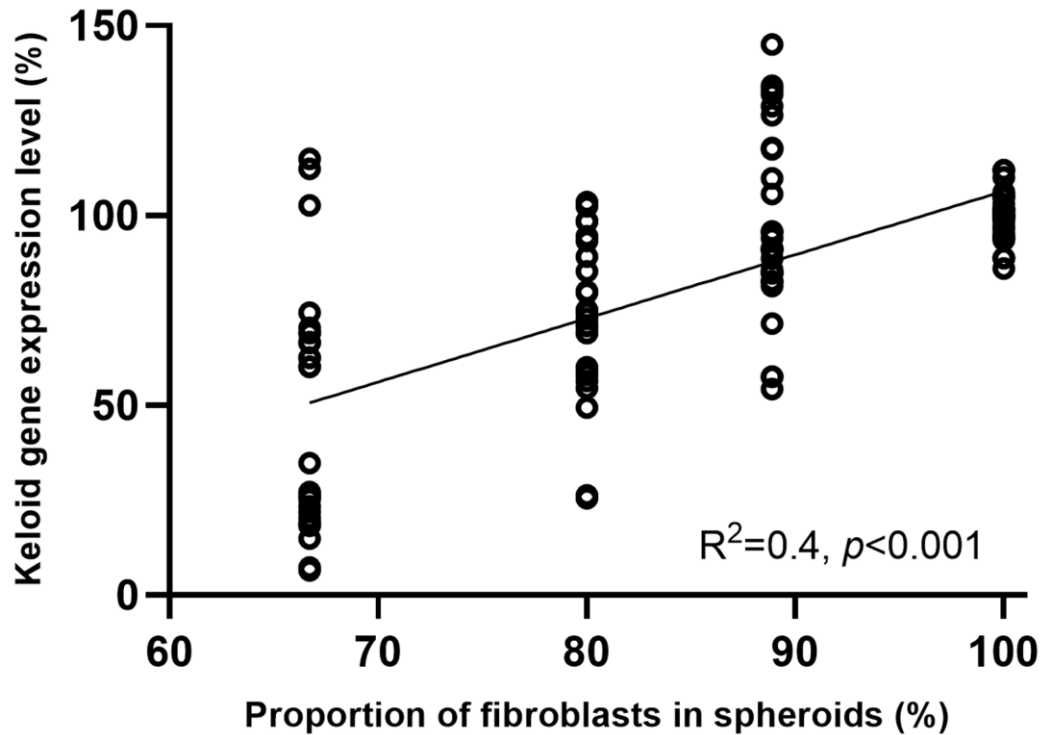


Supplementary Information



Supplementary Figure S1. Simple linear regression analysis demonstrating a positive correlation between the proportion of fibroblasts in spheroids and the ratio of keloid gene expression levels. The x-axis represents the proportion of fibroblasts within the spheroids: 66.7% for F2E1, 80% for F4E1, 88.9% for F8E1, and 100% for F1E0. The y-axis indicates the ratio of keloid gene expression levels relative to the expression level in F1E0, which is set to 100%. The linear regression line illustrates a statistically significant trend of increasing keloid gene expression as the proportion of fibroblasts increases, suggesting a direct relationship between fibroblast concentration and keloid gene expression in the spheroids.

Supplementary Table 1. Clinical information of keloid patients

Sample ID	Sex	Race	Age	Site of Sampling
K1	Male	Asian	50	Chest
K2	Female	Asian	55	Chest
K3	Female	Asian	55	Chest

Supplementary Table S2. Sequences of primers used for RT-PCR studies

Gene code	Forward primer (5'-3')	Reverse primer (5'-3')
<i>ACTB</i>	CCGCGAGAAGATGACCCAGA	GATAGCACAGCCTGGATAGCA
<i>COL1A1</i>	CTGAGCCAGCAGATCGAGAA	CATCTTGAGGTCACGGCAGG
<i>COL1A3</i>	TATCGAACACGCAAGGCTGT	AAAAGCAAACAGGGCCAACG
<i>TGFB1</i>	CGGCCTTTCCTGCTTCTCAT	GAGGTCCTTGCGGAAGTCAA
<i>TGFB3</i>	CCCAGCTCTAAGCGGAATGA	TAGCGCTGTTTGGCAATGTG
<i>HIF1A</i>	GCAGAATGCTCAGAGAAAGCG	TAGCTGCATGATCGTCTGGC
<i>MMP14</i>	CCCAACATCTGTGACGGGAA	TTGGTTATTCCTCACCCGCC
<i>HTRA1</i>	AAGTTCCTCACGGAGTCCCA	GCTCTTTGGCTTTGCTGGAC
<i>ADAM12</i>	TCCGGCAAGCAGATAACCAA	TAAACCACAAATCCGGCAGC
<i>CTHRC1</i>	GTGGCTCACTTCGGCTAAAAT	CAATGGGAAGAGGTCCTGAACA